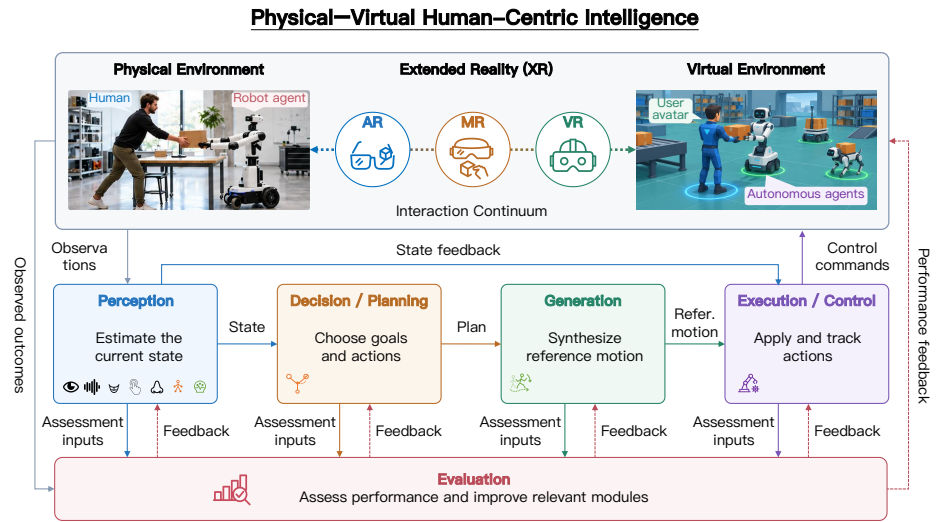


RESEARCH INTERESTS

Focused on human action perception, understanding, and assessment in complex dynamic environments, combining computer vision, multimodal learning, and augmented reality techniques, with emphasis on fine-grained action analysis, cross-modal alignment, and continual learning methods.

I propose a unified physicalvirtual human-centric intelligence framework that places humans, robots, and autonomous agents in an extended-reality interaction continuum. It closes the loop through perception, decision/planning, generation, execution/control, and evaluation, using state feedback, control commands, and performance assessment to coordinate intelligent behavior across environments and support human action understanding and quality assessment in complex dynamic scenes.



WORK EXPERIENCE

 **Postdoctoral Researcher, Tsinghua University**
Department of Psychology and Cognitive Sciences 2025 – Present
Collaborating Advisors: Prof. Jia Liu, Asst. Prof. Liyuan Wang

EDUCATION

 **PhD, Beihang University, Beijing, China**
Computer Application Technology, State Key Lab of VR Technology and Systems 2020 – 2025
Advisor: Prof. Xiaohui Liang

 **Visiting Student, Durham University, Durham, UK** Feb. 2024 – Aug. 2024
Collaborating Advisors: Prof. Hubert P. H. Shum, Dr. Frederick W. B. Li

 **Bachelor, Henan Normal University, Henan, China**
Major: Computer Science and Technology 2016 – 2020
Minor: English 2017 – 2020

 **Exchange Student, Frankfurt University of Applied Sciences, Frankfurt, Germany** June 2019
Collaborating Advisor: Prof. Jürgen Jung

ACADEMIC ACTIVITIES AND SERVICES

➡ Received two postdoctoral research grants:

- China Postdoctoral Science Foundation General Program, 78th Batch (No. 2025M781489)
- National Funded Postdoctoral Researcher Program, Class C, 2026 (No. GZC20260911)

➡ Delivered oral presentations at several top conferences in computer vision and virtual reality, including:

Conference	VR 2025	ECCV 2024	VR 2023	ISMAR 2023	ISMAR 2021
Report Link	v4Yz7dMdM3Q	Llyiw-CpOew	cZbc1ScexMg	D7DL85wP12w	DYrh6KpDKC

➡ Co-supervised multiple Master’s and PhD students in China and UK, including:

Name (YY–YY)	Degree	University	Co-supervisor	Research Field
Ruisheng Han (24–28)	PhD	Durham	Prof. Hubert	Co-authored CVPRW
Ruizhi Cai (22–25)	Master’s	Beihang	Prof. Liang	AQA, co-authored TVCG, IJCAI
Zikai Hao (22–25)	Master’s	Beihang	Prof. Liang	Pose estimation, co-authored TVCG
Chen Chen (21–24)	Master’s	Beihang	Prof. Liang	Hand-object interaction, co-authored ISMAR’23
Zhiyuan Cheng (20–22)	Master’s	Beihang	Prof. Liang	Pose estimation, co-authored ISMAR’21
Yulei Zhong (20–22)	Master’s	Beihang	Prof. Liang	Action recognition

➡ [Main student participant and contributor](#) in an NSFC project (No. 62272019);

➡ Responsible for the [reception and arrangement](#) of Dr. Frederick W. B. Li from Durham University during his visit to Beihang University (June 2023).

➡ Served as a [reviewer](#) for journals and conferences including TPAMI, IJCV, ICML, ICLR, CVPR, and others.

MAIN PUBLICATIONS ([†] DENOTES CO-FIRST AUTHOR; ✉ DENOTES CORRESPONDING AUTHOR)

➡ Among published/accepted papers, 10 are CCF/CAAI A-rank, including 1 ECCV Oral.

- [1] **K. Zhou**, R. Cai, X. Wang, Y. Zheng, L. Wang, J. Li, and X. Liang, “Two-stage multi-modal fusion with adaptive alignment for action quality assessment,” *International Journal of Computer Vision (IJCV)*, vol. 134, no. 9, p. 417, 2026. DOI: [10.1007/s11263-026-02960-7](#). [Online]. Available: <https://arxiv.org/abs/2607.07438>. (🔗 | **CCF A** | **IF10.3**) | 🧠 *Proposes a two-stage multi-modal fusion and adaptive alignment framework that effectively integrates heterogeneous modalities while improving robustness to missing modalities and labels in action quality assessment.*
- [2] **K. Zhou**, H. P. Shum, F. W. Li, X. Zhang, and X. Liang, “Phi: Bridging domain shift in long-term action quality assessment via progressive hierarchical instruction,” *IEEE Transactions on Image Processing (TIP)*, vol. 34, pp. 3718–3732, 2025. DOI: [10.1109/TIP.2025.3574938](#). (🔗 | **CCF A** | **IF13.7**) | 🧠 *Proposing a domain shift...nd feature adaptation outperforms task reconstruction methods.*
- [3] **K. Zhou**, L. Wang, X. Zhang, H. P. Shum, F. W. Li, J. Li, and X. Liang, “Magr: Manifold-aligned graph regularization for continual action quality assessment,” in *European Conference on Computer Vision (ECCV)*, vol. 15069, 2024, pp. 375–392. DOI: [10.1007/978-3-031-73247-8_22](#). (🔗 | 📺 | **CAAI/THU A** | 🏆 **Oral presentation at top computer vision conference, Top 2.3% of conference**) | 🧠 *Proposes continual graph regularization to address the dual challenges of non-stationary generalization and privacy protection in action quality assessment.*
- [4] **K. Zhou**, C. Li, Q. Pan, and L. Wang, “Brima: Bridged modality adaptation for multi-modal continual action quality assessment,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026, pp. 38 904–38 914. [Online]. Available: <https://arxiv.org/abs/2602.19170>. (🔗 | **CCF A**) | 🧠 *Proposes a bridged modality adaptation framework that transfers cross-modal knowledge to mitigate modality absence and catastrophic forgetting in continual action quality assessment.*

- [5] Y. Tao, **K. Zhou**, C. Li, and L. Wang, “Geometry-aware cross-modal graph alignment for referring segmentation in 3D gaussian splatting,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026, pp. 20 433–20 442. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2026/papers/Tao_Geometry-Aware_Cross-Modal_Graph_Alignment_for_Referring_Segmentation_in_3D_Gaussian_CVPR_2026_paper.pdf. (**CCF A**) | Proposes geometry-aware cross-modal graph alignment that jointly models 3D geometric relations and language semantics for referring segmentation in 3D Gaussian splatting.)
- [6] **K. Zhou**, Z. Hao, L. Wang, and X. Liang, “Adaptive score alignment learning for continual perceptual quality assessment of 360-degree videos in virtual reality,” *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, vol. 31, no. 5, pp. 2880–2890, 2025. DOI: [10.1109/TVCG.2025.3549179](https://doi.org/10.1109/TVCG.2025.3549179). (| **CCF A** (**IF6.5**) | **Top-tier journal in computer graphics, oral presentation at top VR conference VR 2025, high-scoring paper recommended for publication in TVCG** | Proposes a continual video quality assessment model based on compressed video replay to address storage and computational constraints on VR devices.)
- [7] **K. Zhou**, R. Cai, Y. Ma, Q. Tan, X. Zhang, J. Li, S. Jin, and X. Liang, “A video-based augmented reality system for human-in-the-loop muscle strength assessment of juvenile dermatomyositis,” *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, vol. 29, no. 5, pp. 2456–2466, 2023. DOI: [10.1109/TVCG.2023.3247092](https://doi.org/10.1109/TVCG.2023.3247092). (| | **CCF A** (**IF6.5**) | **Top-tier journal in computer graphics, oral presentation at top VR conference VR 2023, high-scoring paper recommended for publication in TVCG, Top 10% of conference** | Combines augmented reality to address trust issues in deep learning models for medical rehabilitation. Experiments at Capital Institute of Pediatrics show that the system improved doctors’ diagnostic accuracy by 4.99% and efficiency by 20%.)
- [8] **K. Zhou**, H. P. Shum, F. W. Li, and X. Liang, “Multi-task spatial-temporal graph auto-encoder for hand motion denoising,” *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, vol. 30, no. 1, pp. 6754–6769, 2024. DOI: [10.1109/TVCG.2023.3337868](https://doi.org/10.1109/TVCG.2023.3337868). (| | **CCF A** (**IF6.5**) | **Top journal in computer graphics** | An extension of ISMAR 2021, introducing a motion prediction task to address the oversmoothing problem in denoising results.)
- [9] **K. Zhou**, J. Li, R. Cai, L. Wang, X. Zhang, and X. Liang, “Cofinal: Enhancing action quality assessment with coarse-to-fine instruction alignment,” in *International Joint Conference on Artificial Intelligence (IJCAI)*, 2024, pp. 1771–1779. DOI: [10.24963/ijcai.2024/196](https://doi.org/10.24963/ijcai.2024/196). (**CCF A**) | Proposes a coarse-to-fine task alignment method to address domain shift in action quality assessment.)
- [10] C. Li[†], **K. Zhou**[†], and L. Wang, “Pace: Pretrained audio continual learning,” in *International Conference on Learning Representations (ICLR)*, 2026. arXiv: [2602.03355](https://arxiv.org/abs/2602.03355). [Online]. Available: <https://openreview.net/forum?id=k5PgSlNc4E>. (**CCF A**) | Studies continual learning for pretrained audio models and develops a learning mechanism that balances adaptation to new knowledge with retention of pretrained knowledge.)
- [11] **K. Zhou**, R. Cai, L. Wang, H. P. H. Shum, and X. Liang, “A comprehensive survey of action quality assessment: Method and benchmark,” *Pattern Recognition (PR)*, vol. 179, p. 113 933, 2026. DOI: [10.1016/j.patcog.2026.113933](https://doi.org/10.1016/j.patcog.2026.113933). (| **CCF B** (**CAS 1 TOP**) (**IF9.1**) | Systematically reviews action quality assessment methods, datasets, and evaluation protocols, establishes a unified taxonomy and benchmark, and identifies key challenges and future directions.)
- [12] **K. Zhou**, R. Cai, H. P. H. Shum, F. W. B. Li, and X. Liang, “Neuropath: Brain-inspired dual-pathway graph convolutional networks for skeleton-based action recognition,” *Pattern Recognition (PR)*, vol. 183, p. 114 723, 2026. DOI: [10.1016/j.patcog.2026.114723](https://doi.org/10.1016/j.patcog.2026.114723). [Online]. Available: <https://arxiv.org/abs/2608.17487>. (| **CCF B** (**CAS 1 TOP**) (**IF9.1**) | Proposes a brain-inspired dual-pathway graph convolutional network that decouples and jointly models complementary skeletal motion information for improved action recognition.)
- [13] **K. Zhou**, Y. Ma, H. P. H. Shum, and X. Liang, “Hierarchical graph convolutional networks for action quality assessment,” *IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)*, vol. 33, no. 12, pp. 7749–7763, 2023. DOI: [10.1109/TCSVT.2023.3281413](https://doi.org/10.1109/TCSVT.2023.3281413). (| **CCF B** (**CAS 1 TOP**))

- IF10.8** | Proposes a hierarchical graph network to address the insufficient sensitivity of action quality assessment models to subtle action differences.)
- [14] **K. Zhou**, Z. Cheng, H. P. Shum, F. W. Li, and X. Liang, “Stgae: Spatial-temporal graph auto-encoder for hand motion denoising,” in *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 2021, pp. 41–49. DOI: [10.1109/ISMAR52148.2021.00018](https://doi.org/10.1109/ISMAR52148.2021.00018). (| **Core A*** **CCF B** | **Oral presentation, top-tier conference in augmented reality** | Proposes a spatial-temporal graph model to address motion perception anomalies in augmented reality caused by user fatigue or physiological factors.)
- [15] **K. Zhou**[†], C. Chen[†], Y. Ma, Z. Leng, H. P. H. Shum, F. W. B. Li, and X. Liang, “A mixed reality training system for hand-object interaction in simulated microgravity environments,” in *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 2023, pp. 167–176. DOI: [10.1109/ISMAR59233.2023.00031](https://doi.org/10.1109/ISMAR59233.2023.00031). (| | **Core A*** **CCF B** | **Oral presentation, top-tier conference in augmented reality** | Combines augmented reality technology to propose an object grasping system in simulated microgravity, providing the public with a low-cost, highly immersive space experience tool.)
- ➡ Under Revision:
- [16] **K. Zhou**, Q. Pan, X. Zhang, H. P. H. Shum, F. W. B. Li, X. Liang, and L. Wang, “Continual action quality assessment via adaptive manifold-aligned graph regularization,” *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2026, Under Revision, Round 4. (**CCF A** **IF18.6**) | Proposes adaptive manifold-aligned graph regularization that dynamically models structural relations between previous and new tasks to mitigate catastrophic forgetting and representation drift in continual action quality assessment.)
- [17] **K. Zhou**, Z. Hao, L. Wang, and X. Liang, “Inclusive hand pose estimation via topology-aware distillation and progressive kinematic reasoning,” *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 2026, Under Revision, Round 2. (**CCF A** **IF6.5**) | Proposes an inclusive hand pose estimation framework that combines topology-aware distillation and progressive kinematic reasoning to improve robustness under finger incompleteness while preserving performance on complete hands.)
- ➡ Under Review:
- [18] **K. Zhou**[†], C. Lan[†], D. Li, J. Zhu, and L. Wang, “Proca: Progressive contrastive alignment for robust eeg visual decoding,” *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2026, Under Review. (**CCF A** **IF18.6**) | Proposes a progressive contrastive alignment framework that gradually reduces the cross-modal gap between EEG representations and visual semantics to improve the robustness and generalization of EEG visual decoding.)
- [19] **K. Zhou**, Z. Hao, Z. Chang, L. Wang, and X. Liang, “Palmcl: Structure-decoupled continual learning for hand pose estimation,” in *IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR)*, Under Review, 2027. (**CCF A**)
- [20] H. Yan[†], **K. Zhou**[†], Q. Cheng, W. Dong, C. Lan, J. Zhou, G. Sun, Q. Li, Y. Zhong, and L. Wang, “Brain-inspired hierarchical modularity for general continual learning,” *Nature Machine Intelligence (NMI)*, 2026, Under Review. (**IF29.8**) | Inspired by hierarchical modularity in the *Drosophila* olfactory learning and memory system, proposes a general continual learning framework that separates conflicting experiences while integrating compatible knowledge for robust adaptation to online, blurry, and uncertain data streams.)

MAJOR HONORS AND AWARDS

- 2026.08** Outstanding Doctoral Dissertation Award, Beijing Society of Image and Graphics (1 of 5 recipients)
- 2025.07** Beijing Outstanding Graduate
- 2025.03** No. 9 Company Special Scholarship, Beihang University

(Continued on next page)

2024.11		National Scholarship (PhD)
2024.06		Beihang Academic Innovation Achievement Award
2020.12		Henan Province Outstanding Bachelor's Thesis
2020.12		National Second Prize, China Graduate Mathematical Modeling Competition
2019.11		National Scholarship (Undergraduate)
2019.05		Youth May Fourth Medal, Henan Normal University
2018.12		National First Prize, China Undergraduate Mathematical Contest in Modeling